Status	Finished
Started	Thursday, 27 August 2026, 3:24 PM
Completed	Thursday, 27 August 2026, 3:32 PM
Duration	7 mins 46 secs
Marks	13.00/13.00
Grade	**16.00** out of 16.00 (**100**%)

Question 1

Complete

Mark 1.00 out of 1.00

What is a major disadvantage of **RZ encoding**?

a. It provides very little timing information for receiver synchronization.

b. It uses only positive voltage levels for all transmitted bits.

c. It requires more signal transitions and therefore uses more bandwidth.

d. It always generates a large DC component during transmission.

e. It cannot represent both binary 0 and binary 1 correctly.

The correct answer is: It requires more signal transitions and therefore uses more bandwidth.

Question 2

Complete

Mark 1.00 out of 1.00

Which feature gives Manchester encoding its self-synchronization ability?

a. A transition at the middle of every bit interval

b. Four voltage levels

c. A zero-voltage level between every two bits

d. Variable bit duration

e. A DC component

The correct answer is: A transition at the middle of every bit interval

Question 3

Complete

Mark 1.00 out of 1.00

Which statement correctly compares **Manchester encoding** and **Differential Manchester encoding**?

- a. Manchester has no required transition within each bit, while Differential Manchester always returns to zero within each bit.
- b. Manchester uses the mid-bit transition for both data and synchronization, while Differential Manchester uses the mid-bit transition only for synchronization and the beginning transition for data.
- c. Manchester uses three voltage levels for transmission, while Differential Manchester uses only two fixed voltage levels.
- d. Manchester represents data using fixed voltage levels, while Differential Manchester represents data using zero-voltage intervals.
- e. Manchester uses the beginning transition for both data and synchronization, while Differential Manchester uses only the mid-bit transition for data.

The correct answer is: Manchester uses the mid-bit transition for both data and synchronization, while Differential Manchester uses the mid-bit transition only for synchronization and the beginning transition for data.

Question 4

Complete

Mark 1.00 out of 1.00

What is the main purpose of adding **redundancy in block coding**?

- a. To convert each 5-bit codeword back into a shorter 4-bit sequence
- b. To transform digital data into an analog signal before transmission
- c. To create longer sequences of consecutive zeros in the transmitted data
- d. To remove all signal transitions from the transmitted digital waveform
- e. To improve synchronization and increase the reliability of line coding

The correct answer is: To improve synchronization and increase the reliability of line coding

Question 5

Complete

Mark 1.00 out of 1.00

Which statement correctly describes **8B/6T encoding**?

- a. It represents each 6-bit data block using eight different voltage levels.
- b. It encodes six input bits into eight Manchester-coded signal elements.
- c. It represents each 8-bit data block using six symbols with three voltage levels.
- d. It converts eight analog samples into six binary data elements.
- e. It replaces each 8-bit data block with six binary signal elements.

The correct answer is: It represents each 8-bit data block using six symbols with three voltage levels.

Question 6

Complete

Mark 1.00 out of 1.00

What is an important advantage of representing information in **digital form**?

- a. It eliminates the need to encode data before sending the signal.
- b. It removes the need to sample an analog signal before conversion.
- c. It provides better tolerance to noise during signal transmission.
- d. It requires an unlimited number of signal levels for representation.
- e. It allows transmission without requiring any channel bandwidth.

The correct answer is: It provides better tolerance to noise during signal transmission.

Question 7

Complete

Mark 1.00 out of 1.00

Which sequence correctly describes basic PCM conversion?

- a. Encoding → quantization → sampling
- b. Sampling → quantization → encoding
- c. Line coding → modulation → sampling
- d. Quantization → sampling → amplification
- e. Encoding → sampling → modulation

The correct answer is: Sampling → quantization → encoding

Question 8

Complete

Mark 1.00 out of 1.00

Which statement is **NOT correct** about the process of converting an analog signal into digital form?

- a. Quantization maps the sampled amplitudes into a finite set of discrete levels.
- b. Sampling selects signal values from the original analog waveform at specific time intervals.
- c. PAM produces a sequence of pulses whose amplitudes follow the sampled analog signal.
- d. PAM directly converts each sampled amplitude into its final binary bit sequence.
- e. Quantized values can be represented using binary codes for digital transmission.

The correct answer is: PAM directly converts each sampled amplitude into its final binary bit sequence.

Question 9

Complete

Mark 2.00 out of 2.00

An analog signal has a highest frequency of **6 kHz** and each sample is represented by **8 bits**.

The minimum sampling rate is 12 Ksamples/S,

and the resulting PCM bit rate is 96 Kbps.

Question 10

Complete

Mark 3.00 out of 3.00

An analog signal has a maximum frequency of **10 kHz** and is quantized into **256 levels**.

- The minimum sampling rate is 20 Ksamples/S.
- Each sample requires 8 bits.
- Therefore, the PCM bit rate is 160 Kbps.